DATASCI 406 FA 26, Quiz 3.2

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Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

In a sample of lead levels, several samples are tied at the median, and the jackknife SE of the median comes out to exactly 0. What does this tell you?

1. The median is estimated without error.
2. The leave-one-out medians take only a few distinct values.
3. The sample is too small. With more samples the jackknife SE would converge to the true SE of the median.
4. The factor $(n-1)/n$ should be left out for the median.

Solution

Answer: 2. Leaving out one observation moves the median at most to a neighboring order statistic, so the $T_{(i)}$ take only two or three values. With ties at the median, those values are all equal and the jackknife SE is 0. Option 1 is wrong because the median does vary from sample to sample. Option 3 is wrong because the ratio of the jackknife SE to the true SE does not converge to 1 as n grows. Option 4 is wrong because the factor scales the SE and cannot make it nonzero. Bootstrap the median instead.

Question 2 (1 point)

The bootstrap-t interval uses the replicates

$$W_b^* = \frac{T_b^* - T}{\widehat{\mathrm{SE}}_b^*}, \quad b = 1, \ldots, B.$$

Which SE goes in the denominator?

1. The jackknife SE from the original sample, the same for every b.
2. The standard deviation of all B bootstrap replicates $T_1^*, \ldots, T_B^*$.
3. The bootstrap SE from the original sample, the same for every b.
4. The jackknife SE computed from bootstrap sample b itself.

Solution

Answer: 4. Each W_b^* is studentized with an SE from its own bootstrap sample, here a jackknife on the n values in sample b, at a cost of $B \times n$ evaluations. Options 1, 2 and 3 all divide every replicate by the same constant, which only rescales $T_b^* - T$. The SE from the original sample is used once, outside W^*, in $\left[T - \widehat{\mathrm{SE}}\, W_{1-\alpha/2}^*,\ T - \widehat{\mathrm{SE}}\, W_{\alpha/2}^*\right]$.

Question 3 (1 point)

Polynomials of degree $d = 1, \ldots, 10$ are fit to the same data. The training MSE falls with every increase in d. Why does this not show that $d = 10$ predicts best?

1. Each fit is scored on the same points that were used to fit it.
2. The training MSE has Monte Carlo error, so small differences are noise.
3. A model with more parameters always predicts new points worse.
4. Leave-one-out MSE always chooses $d = 1$.

Solution

Answer: 1. A flexible model can bend toward each training point, which lowers the training error without lowering the error on a new point. Leaving out the point being predicted removes that advantage. Option 2 is wrong because

nothing in a training MSE is random once the data are fixed. Option 3 is wrong because extra parameters can help or hurt, which is what cross-validation is for. Option 4 is wrong because cross-validation can choose any degree.

Question 4 (1 point)

A bike share predicts tomorrow's rentals y with $\hat{y}$ to decide how many bikes to put out. Running out of bikes is a much worse mistake than leaving some unused. With $e = y - \hat{y}$, three losses are

$$\text{squared: } e^2, \qquad \text{absolute: } |e|, \qquad \text{pinball: } L_\tau(e) = \begin{cases} \tau\, e & e > 0, \\ (1-\tau)(-e) & e \le 0. \end{cases}$$

Which loss should cross-validation use?

1. Squared error.
2. Absolute error.
3. Pinball loss with $\tau = 0.9$.
4. Pinball loss with $\tau = 0.1$.

Solution

Answer: 3. Running out means the prediction was too low, so $e > 0$, and the pinball loss charges $\tau\, e$ for those errors. A large τ makes them the expensive ones: with $\tau = 0.9$, an error of one bike too few costs 0.9 and one bike too many costs 0.1. Option 4 reverses the costs. Options 1 and 2 treat the two directions equally.

Rubric. 1 pt per question, no partial credit.